

Tight probabilistic-frame couplings: a literature-implied answer to Problem 1(d)

Source arXiv:1108.2169; supporting arXiv:2501.02602

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Status: literature-implied answer, complete for Problem 1(d). This is not claimed as a new result. The supporting authors cite the source but do not explicitly identify the implication below.

Source question

Ehler and Okoudjou ask in Problem 1(d), PDF page 9 of [1], whether the W_2 -minimizing coupling of two tight probabilistic frames on $\mathbb{R}^N$, at least one having zero mean, is itself a tight probabilistic frame on $\mathbb{R}^{2N}$. They also ask whether tight couplings other than the product measure exist.

Decisive later identity

For a coupling $\gamma \in \Gamma(\mu, \nu)$, Chen and Schmoll write its frame operator in Section 4.6, equation (4.7), PDF page 13 of [2], as

$$S_\gamma = \begin{pmatrix} S_\mu & \Psi_\gamma \\ \Psi_\gamma^\top & S_\nu \end{pmatrix}, \quad \Psi_\gamma = \int_{\mathbb{R}^N \times \mathbb{R}^N} xy^\top d\gamma(x, y). \quad (1)$$

They also express the transport objective through $\text{tr } \Psi_\gamma$ and observe that graph couplings are never probabilistic frames.

Suppose μ and ν are tight with frame bounds A and B . Then $S_\mu = AI_N$ and $S_\nu = BI_N$, so (1) immediately gives the complete criterion

$$\gamma \text{ is tight on } \mathbb{R}^{2N} \iff A = B \text{ and } \Psi_\gamma = 0. \quad (2)$$

In particular, the equal-frame-bound condition omitted from the wording of Problem 1(d) is necessary. When $A = B$ and one marginal has zero mean, the product coupling has $\Psi_{\mu \otimes \nu} = (\int x d\mu)(\int y d\nu)^\top = 0$, recovering the intended product example.

Answers to both questions

The optimal plan need not be tight. Let $\mu = \nu$ be any nonzero zero-mean tight probabilistic frame with bound $A > 0$. The unique W_2 -optimal plan has cost zero and is the diagonal coupling $\gamma_\Delta = (\text{Id}, \text{Id})_\# \mu$. Its cross block is

$$\Psi_{\gamma_\Delta} = \int xx^\top d\mu(x) = AI_N \neq 0.$$

Thus (2) shows that γ_Δ is not tight. Indeed, its support lies in the proper diagonal subspace, so it is not even a probabilistic frame on $\mathbb{R}^{2N}$. This answers the first question negatively under the intended equal-bound and zero-mean hypotheses.

Nonproduct tight couplings do exist. Fix $0 < \varepsilon < 1$ and let

$$\mu = \nu = \varepsilon\delta_0 + \frac{1-\varepsilon}{2N} \sum_{i=1}^N (\delta_{e_i} + \delta_{-e_i}).$$

This is zero mean and tight with bound $(1-\varepsilon)/N$. Define

$$\gamma = \tfrac{1}{2}(\text{Id}, \text{Id})_{\#}\mu + \tfrac{1}{2}(\text{Id}, -\text{Id})_{\#}\mu.$$

Both marginals equal μ , while the two cross blocks cancel, so $\Psi_\gamma = 0$ and (2) makes γ tight. It is not the product coupling: $\gamma(\{(0,0)\}) = \varepsilon$, whereas $(\mu \otimes \mu)(\{(0,0)\}) = \varepsilon^2$. This works in every dimension.

Scope and search status

A bounded search on 12 August 2026 used the exact question, the source arXiv id, and combinations of “probabilistic tight frame”, “coupling”, “product measure”, “cross moment”, and “Wasserstein minimizer”. Equation (4.7) of [2] was the decisive later result. Its relation to Problem 1(d) is an agent-identified implication, so the provenance is `literature_implied_answer`, not `literature_already_answered`. Problems 1(a)–(c) and the separate POVM problem are outside this note.

References

- [1] M. Ehler and K. A. Okoudjou, *Probabilistic frames: An overview*, arXiv:1108.2169 (2011), Problem 1(d), PDF p. 9.
- [2] D. Chen and M. Schmoll, *Probabilistic frames and Wasserstein distances*, arXiv:2501.02602 (2025), Section 4.6, equation (4.7), PDF p. 13.